CODE REVIEW · AUTOMATED CO-PO ATTAINMENT ENGINE

Review: AI question paper validation

For Mayuri · from Rohan · 28 Oct 2026 · 4 commits on `main` (`2338b04` to `8cedd7a` , 11 files, +1,161)

> **Status: accepted. Nothing blocking.** `POST /question-paper/validate` matches API_CONTRACT §10.7.8 field for field. It checks the AI's answer the way the contract asks and is protected by the internal token. The five items below are small follow-ups for whenever you next touch the service.

HOW IT WAS CHECKED

I compared the request and response models with contract §10.6 and §10.7.8, and read the pipeline and route. I ran the whole test suite in a clean environment: **82 passed**, plus 2 live Gemini tests skipped (they run only with `LIVE_GEMINI=1`).

What works well

- The request (`QpValidateRequest`) and response (`QpAiAnalysis`) use exactly the contract's camelCase fields and enum values.
- The AI's answer must cover every sub-question exactly once, with no duplicates, and the results come back in paper order.
- A `suggestedCoId` that isn't one of the ticked COs, or that equals the CO already entered, is dropped.
- If the AI's answer breaks the format, the service asks once more, then returns 502 `AI_SERVICE_UNAVAILABLE` , as §10.7.8 asks.
- MCQ options are checked (count, no blank or duplicate keys). "Out of syllabus" is flagged only when the syllabus actually supports it.
- The endpoint is protected by the same `X-Internal-Token` as the CAM endpoints.
- The new root `.gitignore` keeps `.env` , `.venv` , `target/` and `.idea` out of git.

Small follow-ups

CONTRACT **1. One live test on the sample paper**

Contract §10.7.9 asks for one live test on the Network Security sample paper. `tests/test_live_gemini.py` covers only the CAM. Add a QP test behind the same `LIVE_GEMINI=1` switch that checks the response parses as `QpAiAnalysis` and has one entry per sub-question.

TESTS **2. A test that calls the route**

The new tests call the pipeline directly. Add a few `TestClient` calls to `/question-paper/validate` with the LLM mocked:

- the call is refused without the token;
- a malformed request gets the `ApiError` envelope;
- a good call returns `success: true` with camelCase fields.

CLEAN-UP **3. Startup warning mentions only the CAM**

`app/main.py:29` warns that "the /cam endpoints accept calls without a token". Without a token, `/question-paper` is now open too. Reword the warning to cover all internal endpoints.

CLEAN-UP **4. Confirm the model name**

`.env.example` now says `LLM_MODEL=gemini-3.5-flash-lite`, while the default in `app/core/config.py:10` is still `gemini-2.5-flash`. Confirm that the new name works with the team's API key, then use the same value in both places.

DOCS **5. Service README**

The service README doesn't mention the new endpoint yet. Add it, with a sample request and response next to the CAM ones.

What comes next

- **Scope change:** AI question generation and the knowledge base were removed from the product on 28 Oct 2026 (contract v2.8, §11), so they are no longer your next piece. The next AI work will be agreed with Rohan.
- Routing between the frontend, Spring Boot and FastAPI (open point O-6): agree with Abhijeet whether Spring Boot proxies the AI calls. Your endpoint paths stay the same either way.
- With Abhijeet: once he builds QPV-3, run a full validation of the sample paper from Spring Boot through your service.

Contract references are to `frontend/API_CONTRACT.md` v2.7.